Visualisations

Predicting Financial Wellbeing Among Kenyan Adults
DataSprint 2026 Multiclass classification of financial status

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Part A — Analysis & EDA Visualisations

Exploratory data analysis and model diagnostics generated in the analysis notebook (index.ipynb).

FIGURE 01

Distribution of Financial Status (Target Variable)

The three-class target. 52.6% of respondents report their situation worsened, making this an imbalanced classification problem (weighted F1 is the metric).

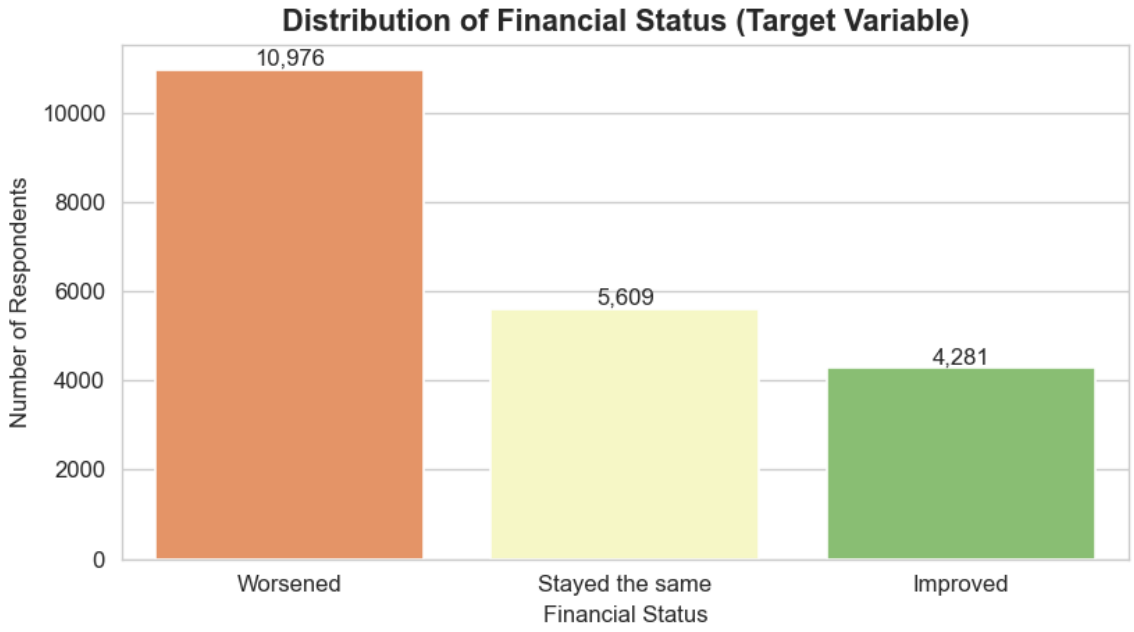


FIGURE 02

Financial Shock vs Financial Status

Respondents who experienced a financial shock are more likely to report 'Worsened' (60% vs 47% for those who did not).

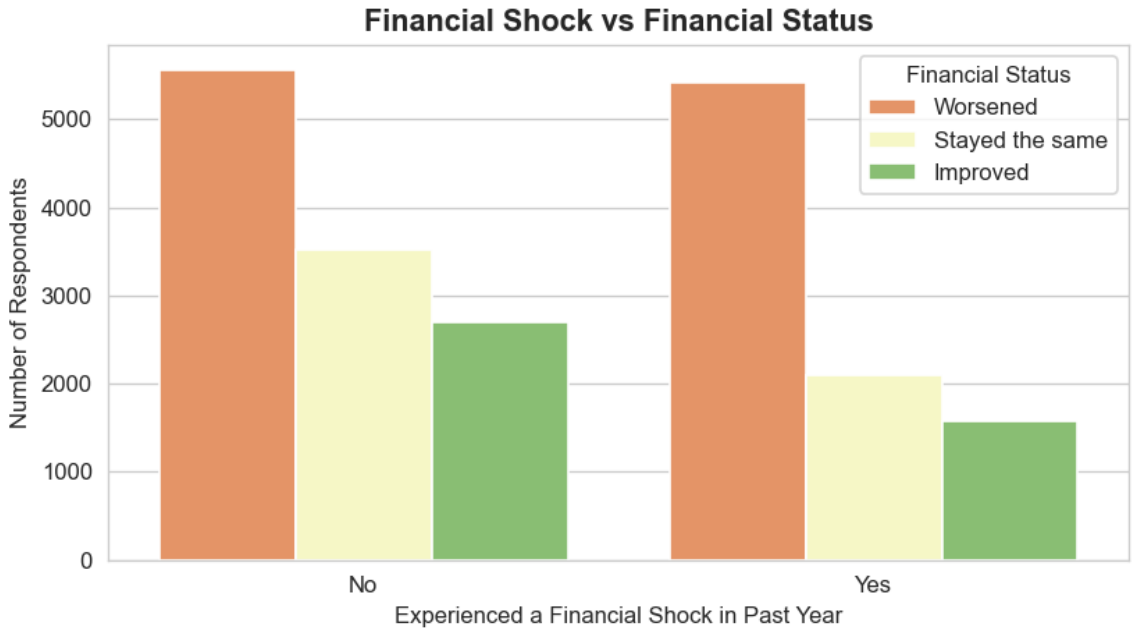


FIGURE 03

Education Level vs Financial Status

'Improved' rises from 12% with no schooling to 33% with university education, but no level is immune. Education alone does not explain who struggles.

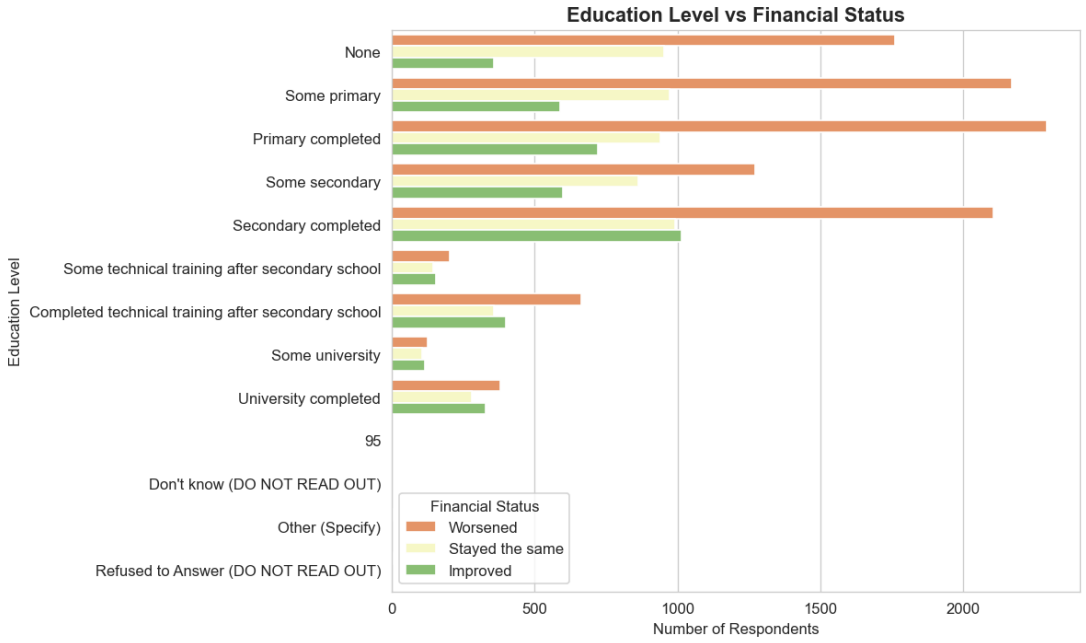


FIGURE 04

Monthly Income Distribution by Financial Status

'Improved' respondents have a higher median income (KES 7,000 vs 5,000), but the wide overlap shows income is far from destiny.

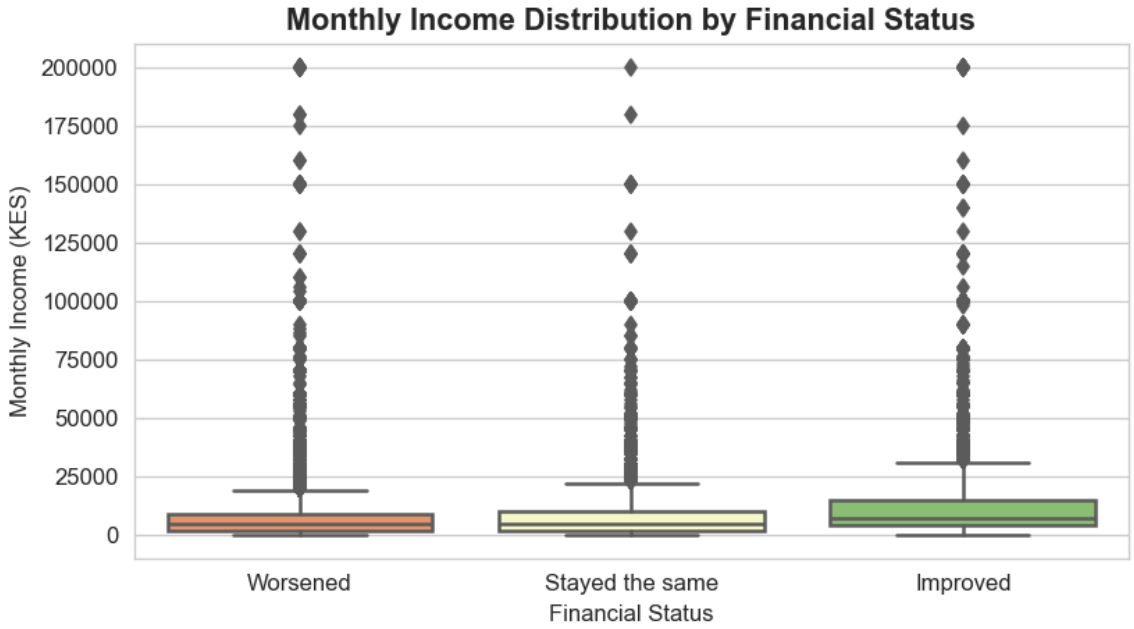


FIGURE 05

Emergency Fund Access (KES 13k) vs Financial Status

Respondents who cannot access KES 13,000 within a month report 'Worsened' more often (59% vs 46%) and 'Improved' half as often (14% vs 28%).

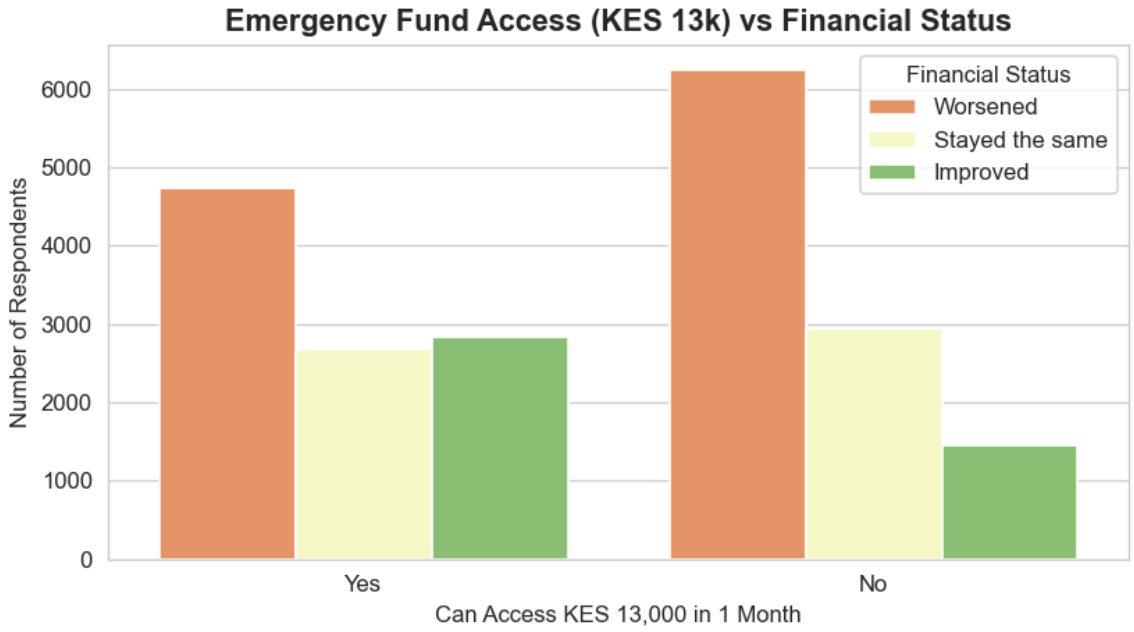


FIGURE 06

Number of Financial Products Used vs Financial Status

More products go with more 'Improved' answers (23% for 3+ products vs 16% for 0-1), but 'Worsened' stays around half in every group.

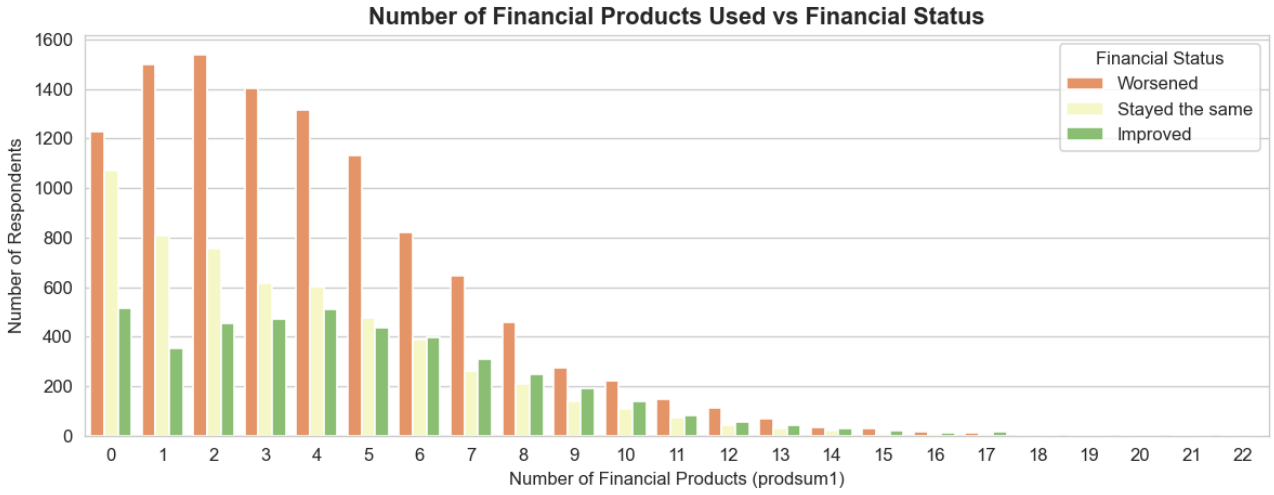


FIGURE 07

Loan Default vs Financial Status

Respondents who defaulted report 'Worsened' more often (58% vs 50%); default is associated with deterioration, in either direction of cause.

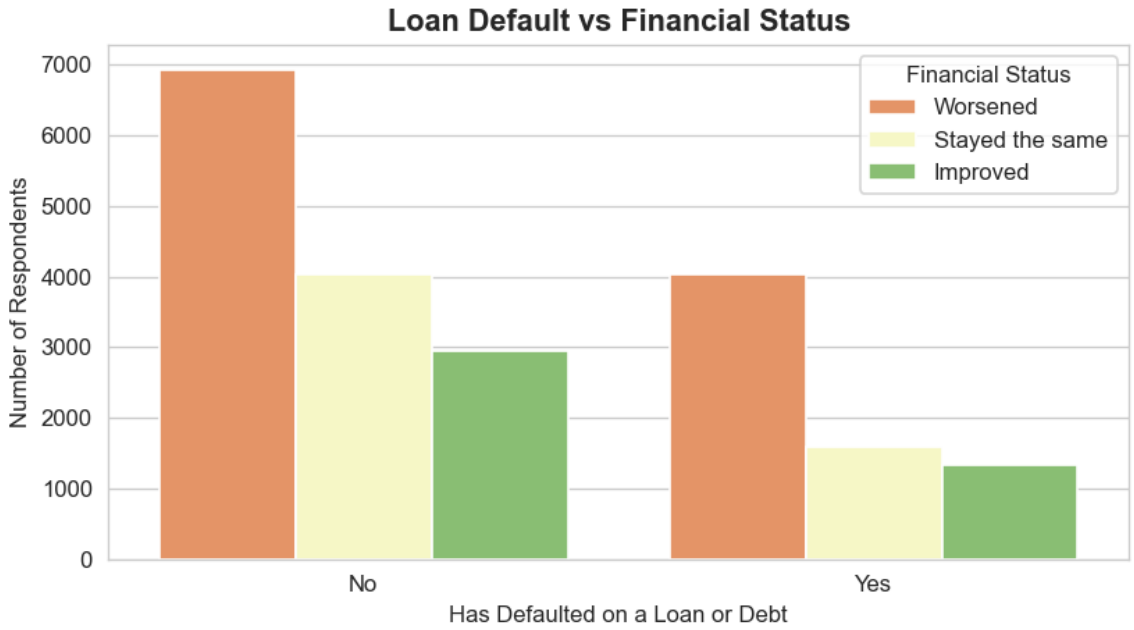


FIGURE 08

Share of Respondents Reporting 'Worsened', by County

Share reporting 'Worsened' by county, from 28% (Wajir) to 73% (Kisumu). Counties have 162-637 respondents each, so small gaps are within sampling noise (about +/-4 points).

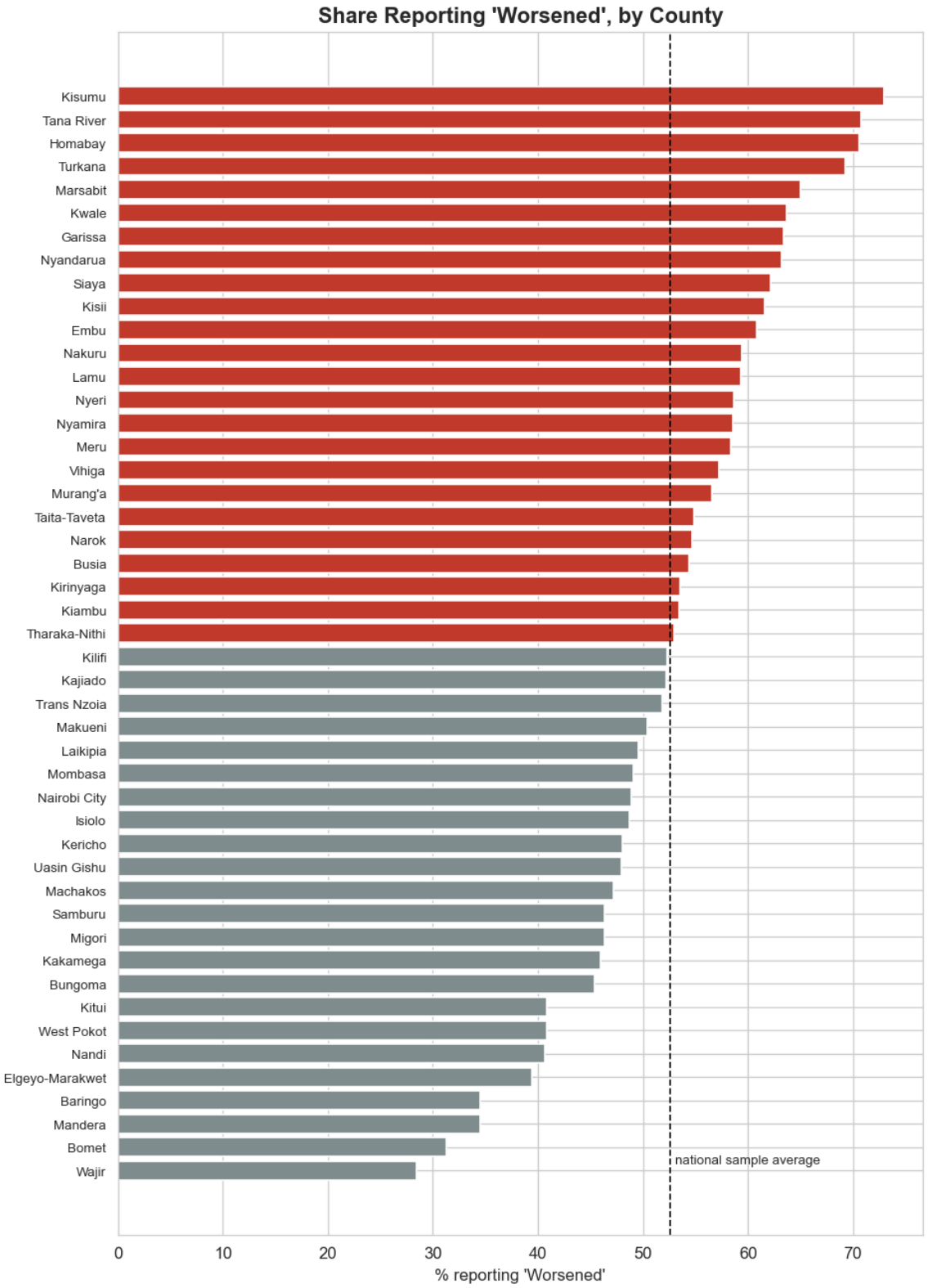


FIGURE 09

Financial Health Score vs Financial Status (Engineered Feature)

Validation of the engineered resilience score: the share reporting 'Worsened' falls steadily from 76% at score 0 to 35% at score 4.

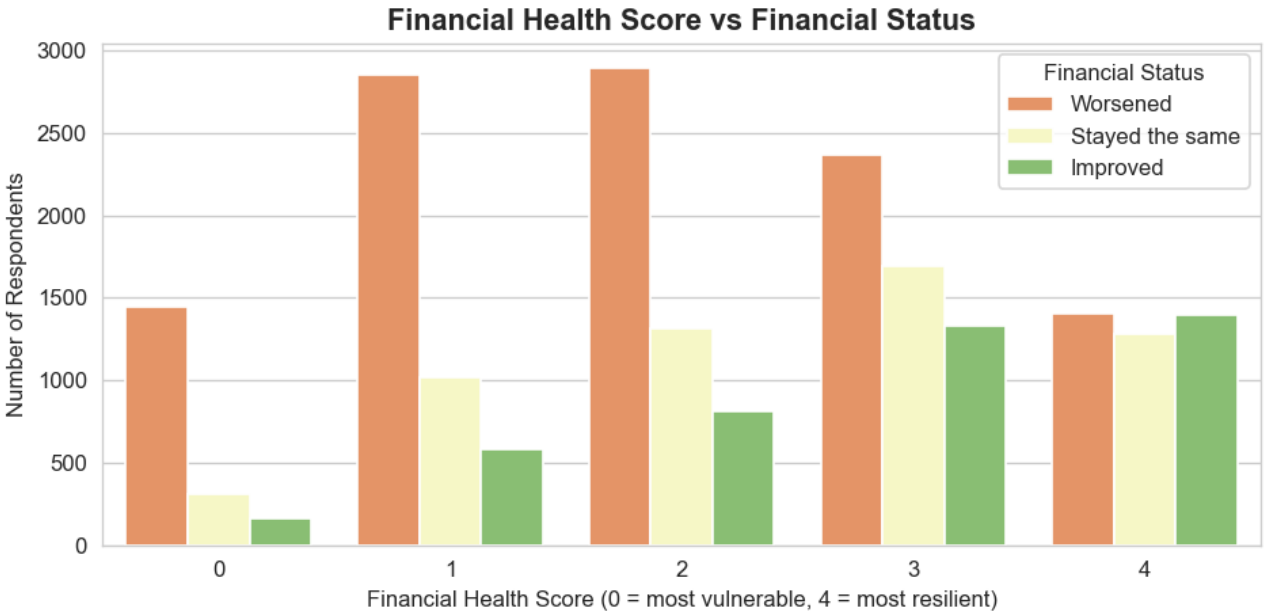


FIGURE 10

Logistic Regression - Confusion Matrix (baseline)

Baseline logistic regression (class-weighted). Confusion matrix on the held-out test set across the three outcome classes.

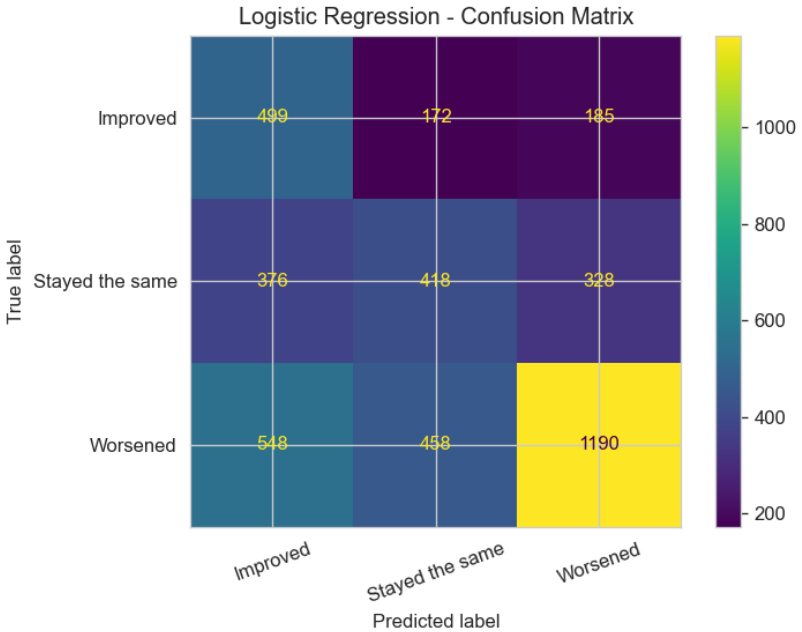


FIGURE 11

Final Model - Confusion Matrix

Final model: LightGBM + SMOTENC with tuned class weights. Strongest on 'Worsened'; 'Stayed the same' is most often mistaken for 'Worsened'.

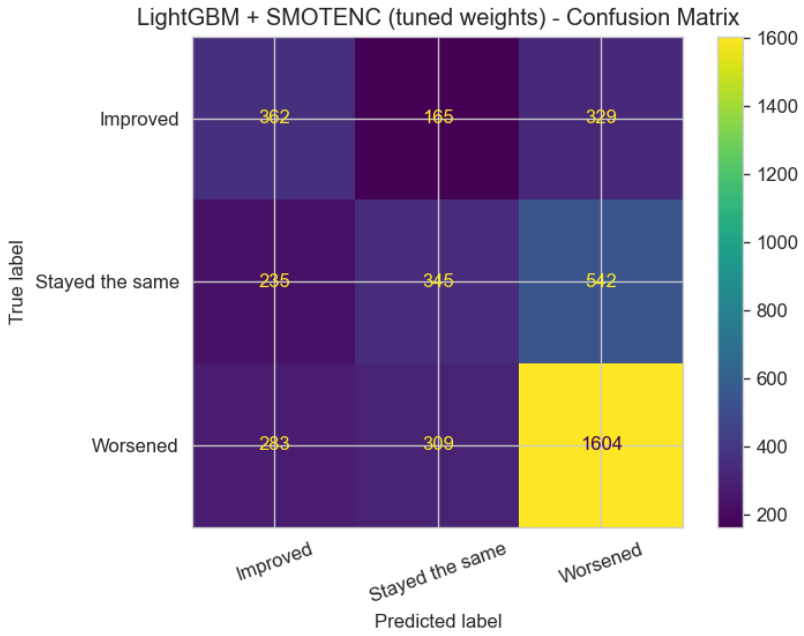


FIGURE 12

Model Comparison - Weighted F1 (CV and Test)

Five models under stratified 5-fold CV (bars, with standard deviation) and on the held-out test set (dots). The model is selected on CV score, not test score.

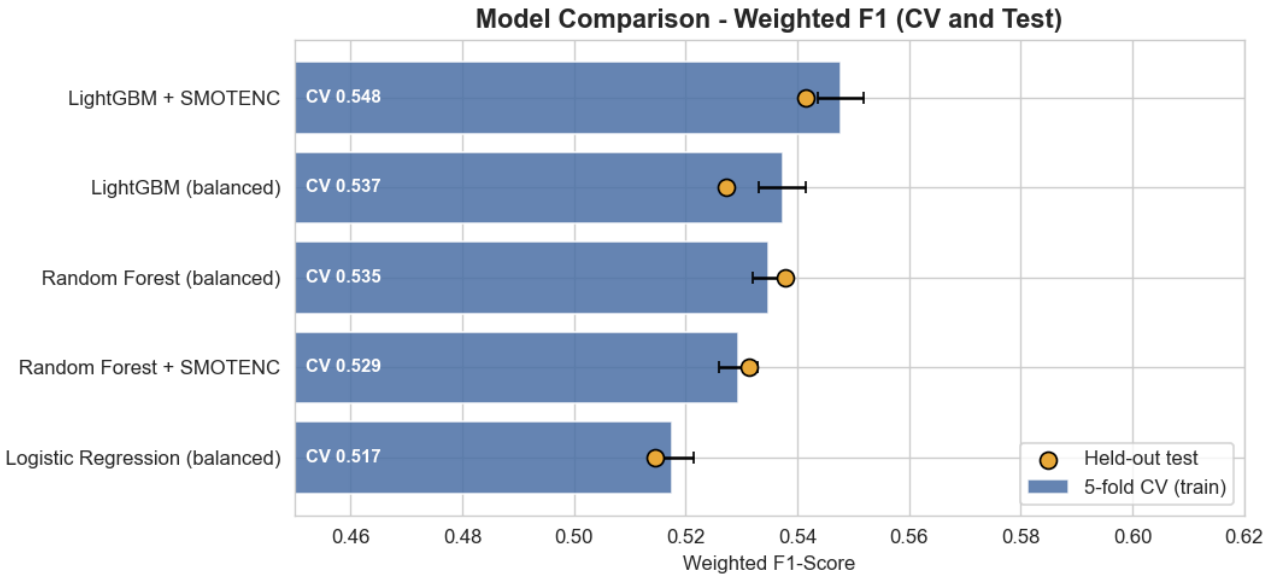


FIGURE 13

Per-Class F1 - Baseline vs Final Model

Per-class F1 on the test set. The final model gains on 'Worsened' but not on the minority classes, where the baseline is slightly higher; weighted F1 rewards the majority class.



FIGURE 14

Robustness - Model Without Self-Reported Resilience Items

Robustness check: refitting without the self-reported resilience items lowers test weighted F1 from 0.541 to 0.520, so the model keeps most of its skill.

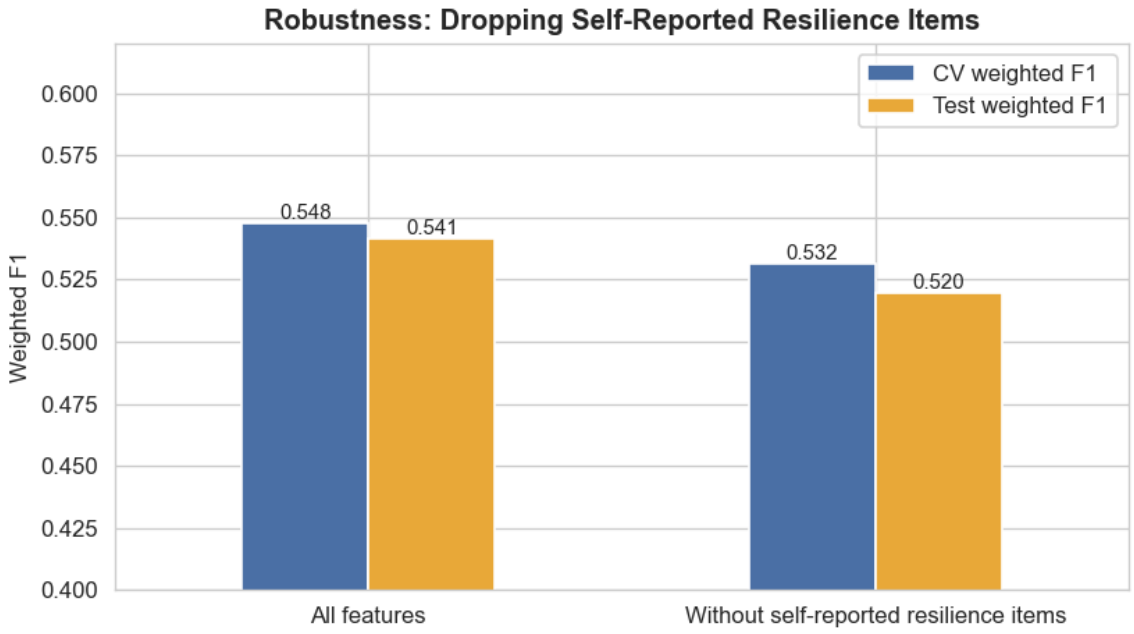
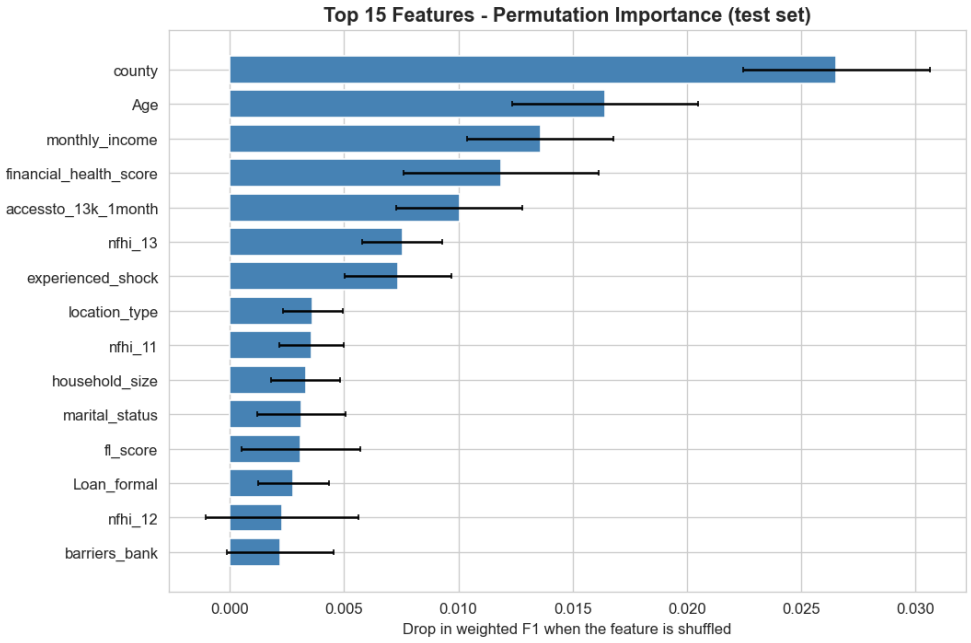


FIGURE 15

Top 15 Features - Permutation Importance

Permutation importance on the test set (drop in weighted F1 when a feature is shuffled). County, age group and income lead, followed by the financial health score and emergency-fund access.



Part B — Presentation Visualisations

Polished, audience-ready charts used in the DataSprint presentation deck.

FIGURE 16

Financial Status One Year On

Presentation view of the target distribution with class percentages.

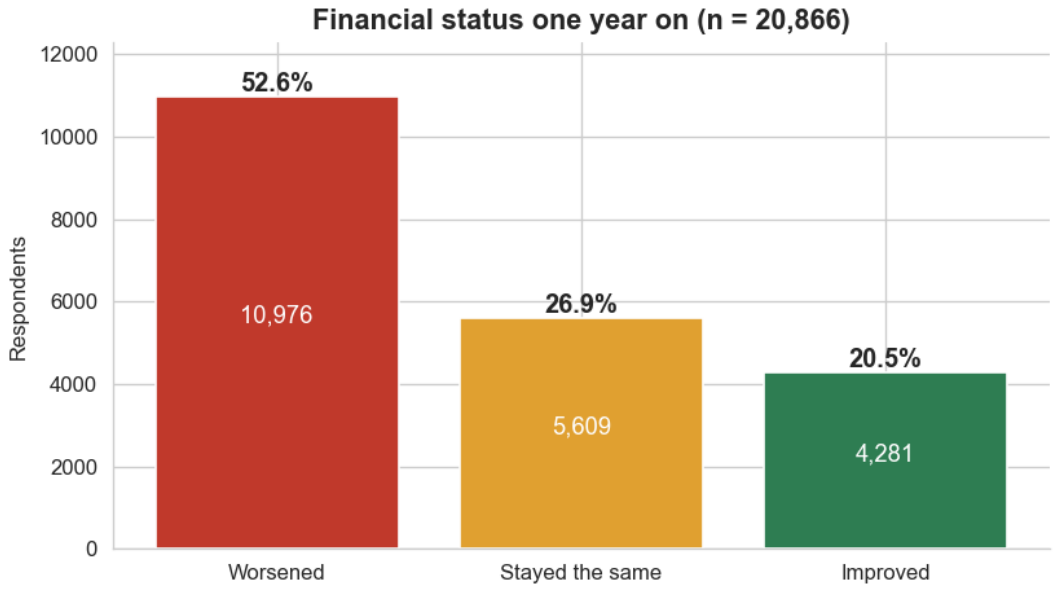


FIGURE 17

Emergency-Fund Access Splits Outcomes Sharply

100% stacked view: respondents who cannot access KES 13,000 are more likely to have worsened (59% vs 46%).

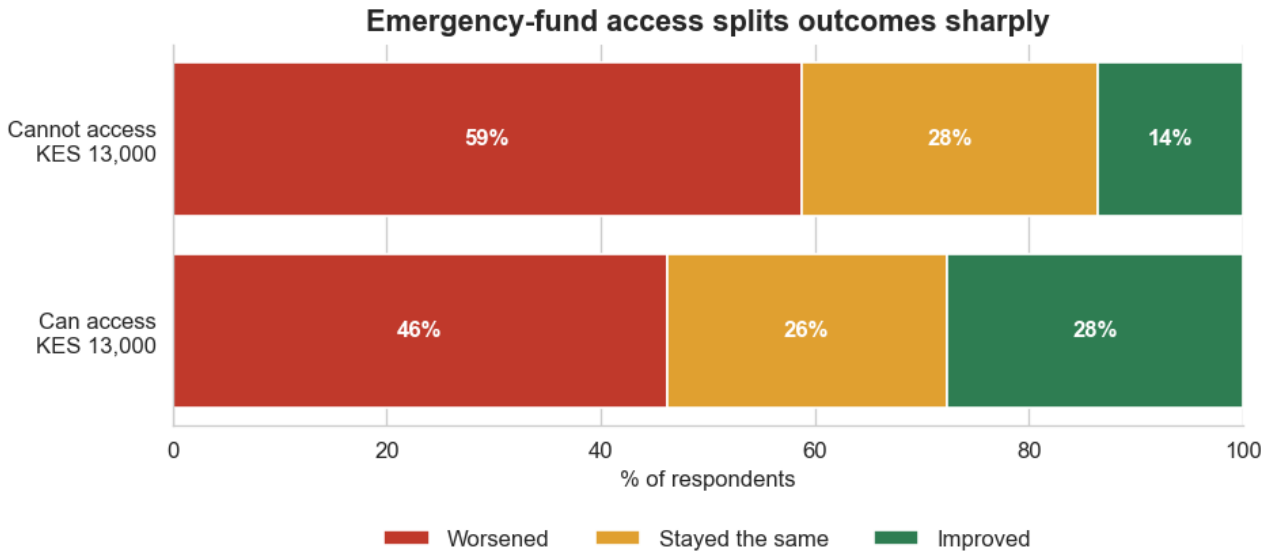


FIGURE 18

Model Comparison - Weighted F1 (Test Set)

Final scoreboard on the test set: LightGBM + SMOTENC with tuned weights (0.542) vs the Logistic Regression baseline (0.515).

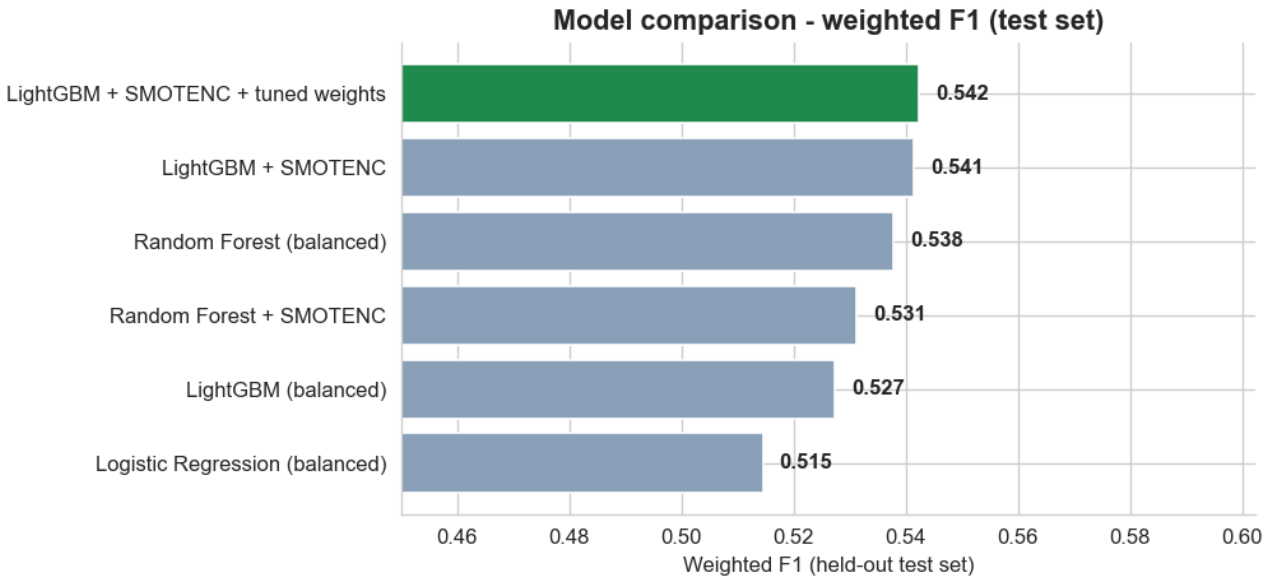


FIGURE 19

Final Model - Confusion Matrix (Test Set, row-normalised)

Presentation-styled confusion matrix (row-normalised) for the selected model on the test set.

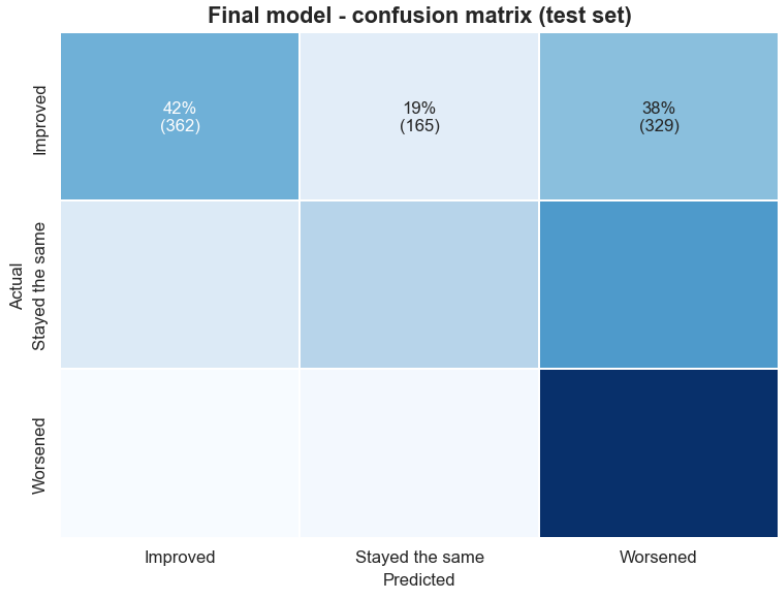


FIGURE 20

Top 10 Drivers of Financial Status

Headline permutation-importance chart used in the deck, with human-readable labels; the engineered feature is highlighted.

